



Generative AI SMART SPEAKER

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Smart speakers such as Alexa, Google Home, and Apple Home have transformed how people interact with technology, enabling voice-based control of smart devices, instant access to information, and efficient task management. However, most still rely

on predefined responses and offer limited intelligence compared to modern generative AI systems.

To overcome these limitations, a Generative AI smart speaker has been developed using advanced large language models (LLMs). Unlike conventional smart speakers, it can generate stories, poems, jokes, emails, and other content on demand while providing more intelligent assistance for everyday

tasks. The system supports multiple AI platforms, enabling seamless switching between providers such as OpenAI, DeepSeek, Grok, and others.

To further extend its capabilities, the smart speaker incorporates a camera. This allows it to read and transcribe books, scan documents, identify objects, describe surroundings, and provide contextual information about items or locations

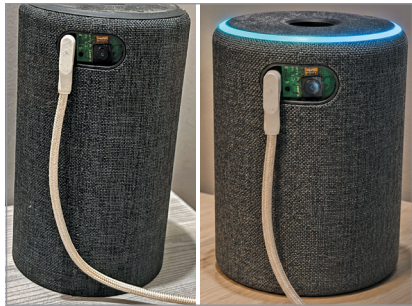


Fig. 1: Author's prototype

```
Defaulting to user installation because normal site-packages is not writeable
Collecting pyporcupine
  Obtaining dependency information for pyporcupine from https://files.pythonhosted.org/packages/ea/88/72d5f
  Downloading pyporcupine-4.0.2-py3-none-any.whl.metadata (5.0 kB)
  Downloading pyporcupine-4.0.2-py3-none-any.whl (3.7 MB)
    3.7/3.7 MB 1.1 MB/s eta 0:00:00
Installing collected packages: pyporcupine
Successfully installed pyporcupine-4.0.2

[notice] A new release of pip is available: 23.2 -> 26.0.1
[notice] To update, run: python3 -m pip install --upgrade pip
```

Fig. 3: Installing Python modules

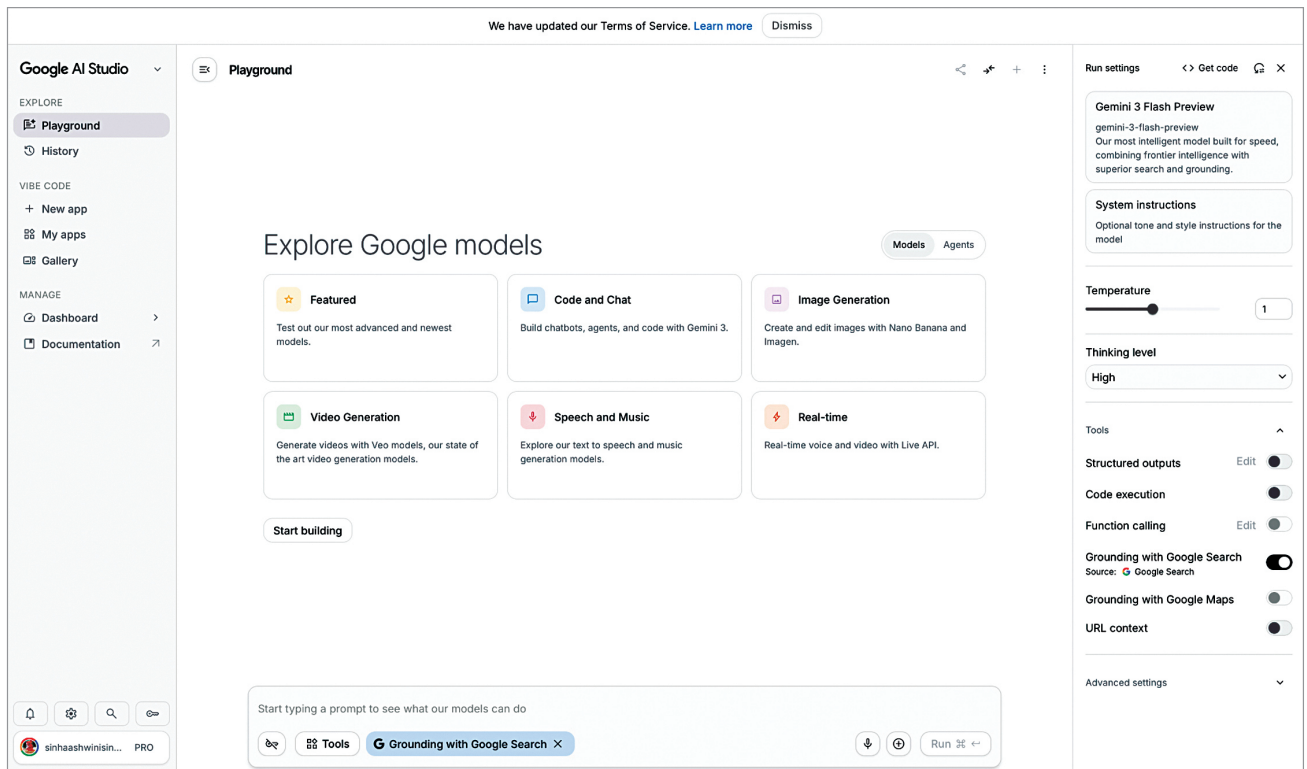


Fig. 2: Creating the Gemini API key